

# Developing an Automated Scoring System for L2 Speaking Proficiency (Spanish Case Study)

Evaluating a second-language speaker's oral proficiency requires assessing multiple aspects of their speech. For an English speaker learning Spanish (or any Romance language), we need a **generalized scoring framework** that can automatically grade their spoken output. Below, we outline key scoring **dimensions**, the **architecture** of an automated evaluation system, and approaches (rule-based and ML-driven) for implementing it. We also draw inspiration from **existing systems** and research to inform best practices.

## Key Scoring Dimensions for Spoken Language

A robust scoring system should break down speaking proficiency into several analytic dimensions (with an option to combine them into an overall score). Common scoring dimensions include <sup>1</sup> :

- **Fluency** – The flow of speech, including rate of speaking, smoothness (few hesitations or pauses), and continuity. Fluent speakers maintain a good pace and minimize long silences or filler words.
- **Pronunciation & Intonation** – Accuracy of producing Spanish sounds (phonemes) and proper word stress/intonation. This covers segmental clarity (e.g. vowel and consonant sounds) and suprasegmental features like stress and rhythm. Good pronunciation ensures the speaker is easily understood.
- **Grammar (Accuracy and Complexity)** – Correct use of Spanish grammar (verb conjugations, gender/number agreement, sentence structure) and the complexity of structures used. High proficiency speakers make few grammatical errors and can use complex sentences when appropriate.
- **Vocabulary (Lexical Resource)** – Range and appropriateness of words used. This includes using varied vocabulary, idiomatic expressions, and words suitable for the context. Advanced speakers use more diverse and precise vocabulary (e.g. beyond basic words).
- **Coherence & Content** – The ability to express ideas in a logical, organized manner. A response should be relevant to the prompt or scenario, with ideas connected clearly. This also covers **task achievement** or relevance: did the speaker address the given scenario or question fully and appropriately?
- **Overall Comprehensibility** – Though somewhat subjective, this reflects how well an average native listener can understand and follow the speaker's response (taking all the above into account).

These dimensions mirror the criteria used in human-scored rubrics like IELTS or CEFR-based exams, which evaluate fluency/coherence, lexical resource, grammatical range/accuracy, and pronunciation for speaking tests <sup>1</sup> . The system can provide **analytic scores** per category (for detailed feedback) and also a **holistic score** summarizing overall proficiency (e.g. a CEFR level or numerical grade). For example, a learner's speaking might be rated *B1 (Intermediate)* overall, with breakdown: Fluency 4/5, Pronunciation 3/5, Vocabulary 4/5, Grammar 3/5, Coherence 4/5.

## System Architecture and Workflow

Any automated speaking evaluation system generally follows a pipeline with three main components <sup>2</sup> :

1. **Speech Recognition (ASR)** – The learner’s spoken response is recorded and fed into an Automatic Speech Recognition engine for transcription. The ASR converts the Spanish speech into text and also provides timing information. A high-accuracy Spanish ASR (potentially a model like Google’s or OpenAI’s Whisper fine-tuned for Spanish) is crucial, since errors in transcription could affect all downstream evaluation.
2. **Feature Extraction & Analysis** – The system analyzes the audio and the ASR transcript to extract relevant features for each scoring dimension:
  3. *Fluency features*: e.g. speech rate (words or syllables per minute), frequency and duration of pauses, and frequency of filled pauses (“um”, “eh”). These can be computed from the timestamped transcript <sup>3</sup> . For instance, one can measure the average length of continuous speech between pauses and the total silent time. More fluent speakers will have a higher speech rate, longer runs of speech between pauses, and fewer/shorter pauses <sup>3</sup> .
  4. *Pronunciation features*: The system can perform a phonetic alignment or use an acoustic model to judge pronunciation quality. A common technique is the **Goodness of Pronunciation (GOP)** algorithm, which scores how closely the learner’s phonemes match a native pronunciation model <sup>4</sup> . Low GOP scores for certain phonemes indicate mispronunciations. Additionally, ASR confidence scores can serve as a proxy – if the ASR struggles to recognize the speech, pronunciation might be unclear. Pronunciation features might include segmental accuracy (phoneme correctness) and prosody (intonation and stress patterns). For example, measuring the variance in pitch and proper stress on syllables can indicate natural intonation.
  5. *Grammar features*: Using the transcript, the system can detect grammatical errors or measure complexity. A rule-based grammar checker (for Spanish) or an NLP model can identify mistakes in verb conjugation, word order, agreement, etc. Complexity can be gauged by sentence length or the use of advanced structures (subordinate clauses, subjunctive mood, etc.). Some automated scoring systems compare the part-of-speech patterns in the response to those in high-proficiency responses <sup>5</sup> <sup>6</sup> – essentially assessing if the grammar usage resembles that of a fluent speaker. For instance, an advanced response might use subordinate clauses (“aunque sea difícil, lo intentaré”) whereas a beginner might use only simple sentences.
  6. *Vocabulary features*: The system can evaluate the richness of vocabulary by measuring the **lexical diversity** (e.g. type-token ratio or number of unique words) and **lexical sophistication**. Sophistication can be assessed by comparing words used to frequency lists or proficiency wordlists – for example, checking if the speaker used higher-level words (less frequent in general use) or mostly common basic vocabulary <sup>7</sup> . If targeting CEFR levels, one could tag words by CEFR level (A1, A2, B1, etc.) and see if the speaker uses any B1/B2-level words. Using more varied and precise words can boost the vocabulary score. However, the **appropriateness** of word choice is also important (using a fancy word incorrectly would not count as true proficiency).
  7. *Coherence and Relevance*: Here the system must analyze the content of the transcript. It might use Natural Language Understanding to check if the response stays on topic and logically addresses the prompt. One approach is to identify key expected points or keywords for the scenario and verify if the learner mentioned them (for scenario-based prompts where certain info is expected). More generally, semantic similarity measures or large language models (LLMs) can be used: e.g.

comparing the learner's answer to an ideal answer or having an LLM judge if the response adequately answered the question. Coherence can also be gauged by the use of discourse markers and logical connectors ("primero..., luego..., por eso..." indicate a well-structured narrative).

8. **Overall factors:** The system might also consider the length of the response (number of words). Extremely short answers may indicate lower proficiency or insufficient development. Many automated tests reward fuller answers: for example, the TOEFL SpeechRater engine noted that top-scoring responses tend to have more words (up to an optimal length) <sup>8</sup>. That said, the content should still be relevant – verbosity alone is not proficiency.

9. **Scoring Model (Evaluation & Scoring)** – In this final stage, the extracted features are fed into a scoring algorithm or model that produces the **scores** for each dimension and/or an overall score. There are two broad approaches here:

10. A **deterministic, rule-based scoring** algorithm can combine the metrics via a formula or rubric. For instance, the system might deduct points for each grammatical error, or assign the fluency score based on thresholds of words-per-minute and pause frequency. A simple rule-based system might say: *If speech rate  $\geq X$  and  $\leq Y$  with  $\leq Z$  long pauses, fluency = "Good"*. Similarly, pronunciation could be scored by the percentage of phonemes above a GOP confidence threshold. This method is transparent and easy to adjust, but it may require careful tuning to align with human-rated standards. One can calibrate it using sample data.
11. A **machine learning scoring model** can learn from data to predict scores more optimally. In this approach, one might train a regression or classification model (e.g. a random forest, SVM, or neural network) using a large set of spoken responses that have been scored by human experts. The model would take all the computed features (fluency, pron, grammar, etc.) as inputs and output a score. Such models can weigh the relative importance of each feature similar to how human raters might (e.g. they might learn that speaking rate and pause length strongly affect fluency, while grammatical errors affect language use scores). For example, ETS's **SpeechRater** uses dozens of features and was found to correlate closely with human scores when calibrated, often within about 0.5 of a human score band <sup>9</sup> <sup>10</sup>. A study notes that an ML-based scoring model (with neural networks using multimodal features) achieved high agreement with human raters on CEFR speaking levels across traits like fluency, accuracy, etc. <sup>11</sup>. The ML approach tends to improve accuracy and consistency, though it requires a **training dataset** of learner responses and can be a "black box" if using complex models.

In practice, many systems use a **hybrid approach**: they rely on an ASR and engineered features, but then use a machine-learned model to map those features to scores <sup>2</sup>. The architecture thus might look like: *Audio -> ASR -> Features -> ML Scoring Model -> Scores*. This pipeline is **fully automated**, enabling instant, unbiased scoring at scale (no human intervention needed) <sup>12</sup> <sup>13</sup>. In fact, modern platforms promise "*AI-powered scoring*" that is consistent and eliminates human rater bias <sup>12</sup>. For example, SpeechAce (a commercial API) analyzes "*pronunciation, fluency, vocabulary, grammar, cohesion and comprehension*" from a speech sample and generates an instant report with sub-scores and feedback <sup>14</sup> <sup>15</sup>.

## Approaches: Rule-Based vs. AI/ML Methods

When building this system, you can consider **rule-based** heuristics, **machine learning** models, or a combination:

- **Rule-Based Evaluation:** In a rule-based design, you explicitly program the scoring logic for each dimension. This requires defining thresholds or error counts that map to score levels. For example:
- **Fluency:** Compute speech rate and pause metrics from the audio. You might say: >120 words/minute = full points for rate, 80–119 = mid score, <80 = low score. Also, if >5 pauses longer than 1 second are present, deduct some points for disfluency. These specific cut-offs can be informed by research – e.g., studies show fluency correlates with fewer and shorter silent pauses <sup>3</sup>. Rule-based fluency scoring might also penalize many self-corrections or repetitions in speech (which indicate hesitation).
- **Pronunciation:** Use a pronunciation confidence measure (like GOP or ASR confidence) for each word or phoneme. For instance, if the average phoneme GOP score is high (meaning the speaker's phonemes largely match native models), pronunciation score is high. Conversely, frequent phoneme-level mismatches or certain hard errors (like mispronouncing **rr** or **j** sounds in Spanish consistently) would lower the score. You could also specifically detect pronunciation of known difficult sounds (e.g. check if “perro” vs “pero” are distinguished). A rule-based approach might set a threshold like “at least 90% of words pronounced clearly (above confidence X) = good pronunciation”.
- **Grammar:** Count grammatical errors in the transcript. For Spanish, one could integrate a grammar-checking library or write rules for common learner errors (such as wrong verb conjugation endings, missing subjunctive where required, incorrect adjective agreement). Each error could subtract from a grammar score. Additionally, measure grammatical complexity: e.g. if the speaker never uses past or future tenses when the scenario demands it, that might indicate limited range. A rule might be: *for full points, the response should include at least one complex sentence or subordinate clause*. This ensures the learner is not just using very basic sentence structures.
- **Vocabulary:** Calculate metrics like the percentage of unique words, and use a Spanish frequency dictionary to gauge word rarity. For example, if the response contains several words that are low-frequency or domain-specific (indicating richer vocabulary), award a higher vocab score <sup>7</sup>. Conversely, if the speaker repeatedly uses a simple word like “bueno” where a more precise term would fit, you might not deduct points, but it could keep the score average. One could also maintain a list of target vocabulary (if a lesson scenario expects certain terms to be used) and check how many were used correctly.
- **Content Relevance:** For scenario-based tasks, you might have an outline of expected content. A rule-based method could look for certain keywords or concepts. For instance, if the prompt is “Describe a past vacation”, the system might expect past-tense verbs (fui, viajé, visité, etc.) and some mention of a location or activity. If the response fails to use past tense or talks off-topic, the relevance/coherence score would drop. This can be done by keyword matching or simpler semantic checks. More advanced: one could encode the prompt and response in vector representations and compute similarity, or require that the response answers all parts of a multi-part prompt.

Rule-based systems are **interpretable** (easy to explain why a score was given) and can be adjusted by domain experts. However, setting the right rules can be challenging. They might be brittle if a learner's response is unexpected but still valid. For example, unusual phrasing might confuse a simplistic grammar

checker but be acceptable Spanish. Therefore, a purely rule-based approach requires extensive refinement and may not capture the full range of speaking ability nuances.

- **Machine Learning and AI Approaches:** Modern systems increasingly use ML or hybrid AI for more robust scoring:
- A classic ML approach is to use the **features** above as input to a trained model. For instance, you gather a dataset of Spanish learners' spoken responses (perhaps from a language test or practice app) with human scores on each dimension. You then train a regression model or classifier that predicts the score given the features (speech rate, errors count, etc.). The model will learn the optimal weighting. For example, it might learn that **pause frequency** and **ASR confidence** are the strongest predictors of fluency score, and assign them more weight than, say, minor filled pauses. This data-driven weighting often outperforms manually set thresholds, and it can handle interactions between features (such as a slight grammar mistake is not as important if vocabulary is very strong, etc., depending on how humans tended to score).
- More advanced is using **deep learning**. Instead of explicitly extracting only a fixed set of hand-crafted features, one can feed the audio and/or transcript into neural network models that learn representations correlated with proficiency. Research has explored end-to-end models (e.g. using transformer or LSTM networks) that input the speech and output a score <sup>16</sup>. These models can automatically learn subtle **prosodic features** (rhythm, tone) and semantic content features. For instance, a model could be trained to predict a CEFR level from the audio by leveraging embeddings from a speech model (like wav2vec or HuBERT for audio, and BERT for text). The cited 2025 study used a transformer-based architecture with advanced prosodic feature extraction to score English speaking, showing the promise of deep learning in this field <sup>17</sup>.
- A practical ML approach without huge data is to use pre-trained **large language models (LLMs)** or similar. One can leverage an LLM (like GPT-4) in two ways: **(a) as a rater** – feed the transcript and ask the model to assess grammar, vocabulary, etc., or give a score based on a rubric; **(b) as a generator** – use it to create varied prompts and even engage in conversation scenarios. Many are beginning to experiment with LLMs for assessment because they have strong language understanding. For example, GPT-4 can *"swiftly evaluate test responses, providing accurate and consistent results, eliminating human bias"* in theory <sup>18</sup>. It can also simulate **conversational assessments**, engaging in a dialogue and then judging the learner's input in context <sup>19</sup>. Using an LLM in evaluation might involve prompt-engineering: you could instruct it with a fixed prompt like "You are a Spanish teacher. Here is a student's response and the expected answer; grade the response on fluency, accuracy, etc. Provide scores 1-5." The model's output could then be parsed as the scores and feedback. Early research has found that GPT-based ratings can correlate reasonably with human ratings, but one must guard against variability – ensuring the prompts are designed to yield consistent criteria. One way to increase consistency is to give the model examples of responses at different skill levels (few-shot prompting) so it calibrates its internal rubric.

Each method has pros and cons. **Rule-based systems** are transparent and can be made deterministic, which might be preferable if you need explainability and control (the user might want to know exactly why they got a certain score). They also don't require large training datasets. However, they might miss nuances and require continuous upkeep for new linguistic phenomena. **ML/AI systems** (especially those using large models) can handle a wider variety of inputs and often align better with human judgment automatically <sup>11</sup>, but they require data and can be opaque. In practice, many successful systems merge these: for example, use deterministic algorithms to get intermediate metrics (like counts, frequencies), and then use an ML model or heuristic formula to combine them into a final score – thereby getting benefits of both.

## Scenario-Based Speaking Exercises

The application envisioned involves **scenario-based** speaking exercises. This means learners are placed in simulated real-life situations (ordering food, job interview, casual conversation, etc.) and must respond in Spanish. This approach is highly engaging and tests the ability to use the language *in context*. Designing a scoring system for scenario-based tasks involves a few additional considerations:

- **Dynamic Prompt Generation:** You can use an AI (like an LLM) to generate varied scenarios and questions. For example, the system might say (text or via a TTS voice): *“Imagine you’re at a restaurant in Madrid and your order was wrong. Please complain to the waiter and resolve the issue.”* The user then responds verbally. AI can help create these prompts on the fly, making the practice more varied and tailored to the learner’s level.
- **Interactive Role-Play:** An “agentic” LM-based system could even play a character in the scenario. It might carry out a dialogue with the learner. For instance, the AI (speaking as the waiter) asks *“¿En qué puedo ayudarle?”*, the learner responds, AI replies, etc. Throughout the interaction, the system is evaluating the learner’s responses. This is akin to what some apps already do: *Speak* (an AI tutor app) and others use OpenAI’s speech recognition and generation to have open-ended conversations with learners <sup>20</sup>. Such **conversational assessments** test not only the learner’s monologue ability but also how well they can *interact* (listen and respond) in the target language.
- **Relevance and Task Completion:** In scenario tasks, an important scoring aspect is **task achievement** – did the learner appropriately fulfill the communicative task? The system needs to judge relevance and completeness of the response. For example, if the scenario is a job interview question and the user’s answer is off-topic or too brief, the score should reflect that. This likely requires semantic analysis. One could program specific criteria for each scenario (e.g., in a “complaint at restaurant” scenario, the ideal response would mention what was wrong with the order and what the person wants as a resolution). The AI can check if those elements were present. Some APIs provide a “relevance” score by comparing the response to the prompt or expected answer <sup>21</sup> <sup>22</sup>. In more open scenarios, using an LLM to judge whether the response is a logical and relevant reaction to the situation can be effective.
- **Soft Skills and Tone:** In professional or social scenarios, aspects like politeness or formality might matter (especially in languages like Spanish with formal vs informal address). An advanced evaluation might consider if the learner used an appropriate register (e.g., using “usted” with a job interviewer or a customer). While this is beyond basic language correctness, it is something an AI could be tuned to notice. For instance, *Bright AI* (an English testing platform) not only evaluates linguistic ability but also soft skills like adaptability and problem-solving in the context of scenario tasks <sup>23</sup>. We might not go that far initially, but it’s good to keep in mind how scenario-based assessment opens doors to evaluating pragmatic skills too.
- **Immersive Experience and Feedback:** Scenario-based tests can feel like a game or real interaction, which is motivating. The system should ideally give **immediate feedback** after the exercise. For example, after the user finishes speaking, the app might display: *“Pronunciation: 7/10 (work on “rr” sound in perro), Fluency: 8/10 (good pace, but a few long pauses), Vocabulary: 6/10 (used appropriate words, but could incorporate more variety), Grammar: 8/10 (minor error in past tense conjugation of ir), Task Completion: 9/10 (addressed the scenario well).”* This granular feedback helps learners know what to improve. Research highlights that real-time feedback, such as pronunciation correction tailored to the user’s L1 interference, makes the experience more effective <sup>24</sup>. The system might even suggest practice exercises targeting the weak areas (e.g., if stress on syllables was a problem, recommend a pronunciation drill on word stress <sup>25</sup>).

In scenario-based practice, **consistency** and **fairness** of scoring remain crucial. The contexts vary, but a B1-level performance in any scenario should get a similar score. AI assessment engines strive for this by using large diverse training data and standardized rubrics, thereby reducing subjective bias. In fact, one advantage of AI-driven scoring is that it can be more objective and “*bias-free*” than human raters (no favoritism, fatigue, or cultural bias) <sup>26</sup> – assuming the system has been trained on diverse data to avoid intrinsic biases.

## Inspiration from Existing Systems

Several existing systems and research projects provide guidance on how to implement our scoring solution:

- **ETS SpeechRater (TOEFL):** This is a mature automated scoring engine for English speaking. It analyzes “critical markers like pronunciation, fluency, vocabulary and grammar” to provide detailed feedback <sup>27</sup>. Under the hood it uses many features; for example, it measures timing (speaking rate, pauses), pronunciation (via acoustic models), vocabulary level (via word frequency analysis), and grammar patterns <sup>28</sup> <sup>7</sup>. It then uses a statistical model to map these to a score. In our Spanish system, we can apply similar features (adjusted for Spanish). ETS research shows the importance of combining features: one study had 12 features for delivery and 6 for language use that together predicted scores well <sup>29</sup> <sup>30</sup>.
- **Pearson Versant:** The Versant tests (available for English and some other languages) use an entirely automated scoring system. Test-takers respond to various prompts (reading aloud, repeating sentences, open questions). The scoring is done by a proprietary speech processing algorithm. While proprietary, Pearson has indicated their “**automated scoring evaluates grammar, vocabulary, pronunciation, and fluency**” <sup>31</sup>. They likely use ASR with pronunciation models and predetermined scoring logic, benchmarked to human raters. One notable aspect of Versant is that it often uses **structured tasks** (like repeat a sentence) which make ASR and evaluation easier (since there’s an expected answer). For our app focusing on free speaking, we have a harder task, but Versant’s success confirms that with careful calibration, automated speaking scores can be highly reliable.
- **SpeechAce API:** This is a commercially available API that developers can integrate to get instant speech scoring. According to their documentation, it can handle multiple languages (English, Spanish, French, etc.) and returns scores on **fluency, pronunciation, grammar, vocabulary, and coherence** for spontaneous speech <sup>1</sup>. It even has specialized endpoints for “relevance” and “task achievement” scoring in open responses <sup>21</sup>. Essentially, SpeechAce encapsulates the pipeline we described: you send it an audio and prompt, and it returns detailed metrics. They report that their scoring aligns closely with standard tests (correlation of ~0.82 with IELTS speaking scores) <sup>9</sup>. While we may build our own system, the existence of SpeechAce and similar services validates the feasibility of an automated Spanish speaking evaluator. We can draw inspiration from their feature set and perhaps use a similar scale (e.g., output scores 0–100 or mapped to CEFR levels).
- **Academic Research:** There is rich research on automated speech scoring. As an example, a 2021 study (de Jong & Pacilly) focused on *automatic fluency measurement*, highlighting features like “*speech rate, frequency/duration of silent pauses, filled pauses (‘eh’, ‘um’), and repetitions/self-corrections*” as objective fluency indicators <sup>3</sup>. Such studies inform which metrics strongly correlate with human judgments of fluency. Another line of research has improved **pronunciation scoring** through methods like GOP and phoneme error detectors, enabling pinpointing of mispronounced sounds <sup>4</sup>. We should leverage these insights: ensure our fluency scoring uses multiple timing features, and pronunciation scoring operates at the phoneme level for detailed feedback.

- **AI-Powered Learning Apps:** Apps like Duolingo, Babbel, and newer AI tutors (e.g., the app *Speak*) are integrating speech evaluation. Duolingo's English Test (for proficiency certification) is an AI-scored test that includes a *speaking* component: test-takers describe an image or respond to a prompt and the AI scores it. They haven't published all details, but presumably they use a combination of ASR and a trained scoring model, similar to ETS and Pearson methods. The key takeaway is that **industry leaders trust automated scoring** for high-stakes assessment, after validating that it correlates well with human scoring. For our purposes (practice and self-improvement), an automated system is even more appropriate, as absolute precision is less critical than giving learners feedback and approximate levels.
- **Bright Language's AI test & Hallo:** As mentioned, *Bright AI* introduced a scenario-based test for workplace English, which "*places test-takers in realistic situations (e.g. job interview) ... delivering consistent, objective scoring and immediate results.*" <sup>32</sup> <sup>13</sup> . This shows the trend towards contextual, open-response testing with AI scoring. Another platform, *Hallo*, offers a five-question speaking test with scenario prompts and uses AI to evaluate the responses <sup>33</sup> . These systems underline the importance of adaptive, realistic tasks with quick feedback. They likely use a mix of ASR and an ML scoring engine behind the scenes. We can mirror this approach for Spanish: use realistic role-play prompts and an AI backend to evaluate the spoken answers instantly and consistently.

To illustrate, modern AI systems can **analyze speech in real time** and give granular feedback. An expert describes that today's AI-driven tools "*do the heavy lifting: scoring pronunciation, tracking fluency, identifying grammatical patterns,*" all automatically <sup>34</sup> . This frees instructors (or the app designers) to focus on using those analytics to help the learner improve. Our app should therefore not only score the learner but also provide **diagnostic feedback**: highlight mispronounced words, suggest correct grammar for mistakes, and perhaps even track progress over time (e.g. "your fluency has improved from 6/10 to 8/10 over the last two weeks").

## Implementation Considerations and Summary

In building this generalized scoring system for Spanish (extendable to other Romance languages), here are some practical considerations:

- **Data and Calibration:** If possible, collect a sample of Spanish learners' recordings with known proficiency labels to calibrate your scoring model. If that's not feasible, start with rule-based heuristics grounded in research (for example, use known CEFR descriptors as a guide: a B2 speaker "can produce speech at a natural speed with little searching for words", which translates to concrete fluency metrics). You can later refine the model with user data.
- **Spanish-Specific Adjustments:** While the general approach works across languages, tweak for Spanish specifics. Spanish ASR is quite reliable due to the phonetic nature of the language, but watch out for regional accents (a learner exposed to Latin American Spanish vs. Castilian might have different pronunciation; the ASR/acoustic model should ideally cover those variations or you might have separate models). For pronunciation scoring, note Spanish phonemes that are challenging for English speakers: rolling **rr**, the **j** ([x] sound), differentiating **b/v** (which are pronounced similarly in Spanish). The system might specifically listen for these; for instance, if a student consistently pronounces "*pero*" and "*perro*" the same, flag it as a pronunciation area. For grammar, typical English-speaker errors include incorrect gender agreement (e.g. *el problema* as feminine) or using indicative when subjunctive is needed – if we can detect those, we can give targeted feedback.



- **Multi-Language Scalability:** Since the goal is a general framework (e.g. to also cover French, Italian, etc.), design the system with language-specific modules. The ASR, pronunciation model, and error rules need to be language-specific. But the overall pipeline remains the same. You might have a core system where you can plug in a language's phoneme set, its POS tagger or grammar checker, and its frequency list. Romance languages share some characteristics (like gendered nouns, similar phonetic inventories), so insights from Spanish will transfer to French or Italian modules fairly well.
- **User Interface:** For the app, ensure the scoring results are presented in a user-friendly way. Perhaps use gauges or bars for each skill, and a concise textual critique. Since language learning is the context, **positive, constructive feedback** is important – highlight what the learner did well (e.g. “Good job using past tense and keeping a steady pace!”) along with what to improve.
- **Alignment with Standards:** If possible, map the scores to a meaningful scale like CEFR or an exam score. SpeechAce, for instance, reports scores aligned to CEFR levels for various languages <sup>35</sup>. You could decide that the overall score 0–100 corresponds to CEFR A1 through C2 bands. This gives learners an idea of their proficiency in standard terms. It can be motivating (e.g. “You’re at a solid B1 level in spoken Spanish”).
- **Continuous Improvement:** An AI-based system can improve over time. As more learners use it, you’ll gather more audio data. This can be used to retrain or fine-tune the scoring model, making it more accurate. Also, user feedback can help identify false negatives/positives (e.g. if many users feel their pronunciation score was unfairly low due to background noise, you might adjust the noise handling).
- **Privacy and Ethics:** Since audio data (potentially containing the user’s voice) is involved, ensure you handle it securely. If using cloud-based AI services, be transparent in user consent. Also, be mindful of any biases: for example, if the system is trained mostly on a certain accent or dialect, it should not unfairly penalize other accents. Including a diverse set of speakers in training (or using robust ASR models) will mitigate this.

In summary, creating a generalized automated speaking evaluation system involves combining **speech technology** with **linguistic expertise**: - We define what to measure (fluency, pronunciation, grammar, vocabulary, coherence). - Use ASR and algorithms to **measure those objectively** (from speech rate to grammar errors). - Employ AI/ML to **aggregate these measures** into reliable scores that emulate human judgment <sup>2</sup>. - And apply it in engaging, scenario-based exercises with instant, tailored feedback to the learner.

By drawing on established systems and research, we can implement an automated Spanish-speaking grader that is both **comprehensive** and **learner-friendly**. It will not only score the learner but also guide them on where to improve, making speaking practice more effective. As AI in language assessment is evolving rapidly, this system can stay up-to-date (for instance, adopting new ASR models or LLM evaluators as they become available) to ensure the evaluations remain accurate and fair. With this approach, evaluating a Spanish learner’s spoken fluency and skills can be done at scale – **instantly, consistently, and adaptively** – empowering learners to improve through continuous feedback <sup>24</sup> <sup>34</sup>.

#### Sources:

1. SpeechAce API Documentation – *Use-cases of SpeechAce (languages, features for fluency, pronunciation, grammar, vocabulary, coherence, etc.)* <sup>1</sup> <sup>21</sup>
2. British Council Assessment Research – *Techniques to measure speech fluency automatically (speech rate, pauses, etc.)* <sup>3</sup>

3. RudderLabs (Medium) – *Goodness of Pronunciation (GOP) as a metric for pronunciation accuracy in CALL systems* <sup>4</sup>
  4. Xu et al. (2021) – *Architecture of a speech auto-marker (ASR, feature extraction, scoring model)* <sup>2</sup>
  5. ETS SpeechRater Service – *Feedback on pronunciation, fluency, vocabulary, grammar for non-native speakers (TOEFL prep)* <sup>27</sup>
  6. TOEFL Resources (Michael Goodine) – *Analysis of SpeechRater features for delivery (fluency/pron) and language use (grammar/vocab) with their impacts* <sup>28</sup> <sup>7</sup>
  7. Bridge Universe (Riel, 2025) – *Overview of AI in language testing, scenario-based assessments (Bright AI), and benefits like bias reduction and instant feedback* <sup>32</sup> <sup>26</sup>
  8. Bridge Universe – *Expert insights on AI providing micro-level performance data: scoring pronunciation, fluency, grammar patterns automatically* <sup>34</sup>
  9. Pipplet Blog (2023) – *How GPT-4 can assist in language testing: automated assessment, simulated conversations, adaptive feedback* <sup>18</sup> <sup>19</sup>
  10. SpeechAce Website – *Product info (automated, unbiased scoring; reports aligned to IELTS/CEFR; immediate results in Spanish, French, etc.)* <sup>35</sup> <sup>15</sup>
-

1 21 22 Use-Cases | Speechace API Documentation - Assess spoken language skills using speech recognition

<https://api-docs.speechace.com/introduction/use-cases>

2 11 The architecture of a speech auto-marker | Download Scientific Diagram

[https://www.researchgate.net/figure/The-architecture-of-a-speech-auto-marker\\_fig1\\_349429966](https://www.researchgate.net/figure/The-architecture-of-a-speech-auto-marker_fig1_349429966)

3 Normal version 1.00

[https://www.britishcouncil.org/sites/default/files/](https://www.britishcouncil.org/sites/default/files/de_jong_and_pacilly_2021_new_techniques_to_measure_fluency_in_speech_automatically_non-technical_summary.pdf)

[de\\_jong\\_and\\_pacilly\\_2021\\_new\\_techniques\\_to\\_measure\\_fluency\\_in\\_speech\\_automatically\\_non-technical\\_summary.pdf](https://www.britishcouncil.org/sites/default/files/de_jong_and_pacilly_2021_new_techniques_to_measure_fluency_in_speech_automatically_non-technical_summary.pdf)

4 End-to-end ASR for phoneme-level pronunciation scoring | Medium

<https://medium.com/@rudderanalytics/a-deep-dive-into-phoneme-level-pronunciation-assessment-b45649db5bb9>

5 6 7 8 28 29 30 How does the TOEFL SpeechRater Work? | Test Resources

<https://www.toeflresources.com/blog/how-does-the-toefl-speechrater-work/>

9 10 12 14 15 35 Speaking & Writing AI Test – Speechace | Pronunciation and fluency assessment via speech recognition

<https://www.speechace.com/speaking-test/>

13 23 24 25 26 32 34 Adaptive, Data-Rich, and Immediate: The New Language Assessment Powered by AI - BridgeUniverse - TEFL Blog, News, Tips & Resources

<https://bridge.edu/tefl/blog/adaptive-data-rich-immediate-new-language-assessment-powered-by-ai/>

16 17 Research on Deep Learning-Based English Speaking Automatic ...

<https://dl.acm.org/doi/pdf/10.1145/3773365.3773381>

18 19 20 How GPT-4 is Revolutionizing Language Testing and Assessment

<https://blog.pipplet.com/how-gpt-4-is-revolutionizing-language-testing-and-assessment>

27 SpeechRater Service: Advanced Spoken-response Scoring Application

<https://www.ets.org/speechrater.html>

31 Automated scoring for speaking tests - Pearson Support

<https://support.pearson.com/languages/s/article/Versant-Automated-scoring-for-speaking-tests>

33 AI that automates language assessment - Hallo

<https://www.hallo.ai/assessment/>